



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 1 Examinations 2025/26

Course Instance Code(s)	4BCT1, 4BP1, 4BP4, 1OA3
Exam(s)	B. Sc. Degree (Computer Science & Information Technology, B. Eng. (Electronic & Computer Engineering) and International Exchange
Module Code(s)	CT417
Module(s)	Software Engineering III
Bonded	No [☒] Yes []
Paper No.	1
External Examiner(s)	Dr. Shoaib Ehsan
Internal Examiner(s)	Dr. Enda Howley *Dr. Effirul Ramlan
<u>Instructions:</u>	Answer any 3 questions. All questions are marked equally.
Duration	2 hours
No. of Pages	8
Discipline(s)	Computer Science
Course Co-ordinator(s)	Dr. Colm O'Riordan

Requirements:

Release in Exam Venue	Yes
MCQ Answersheet	No
Handout	None
Statistical/ Log Tables	None
Cambridge Tables	None
Graph Paper	None
Log Graph Paper	None
Other Materials	None
Graphic material in colour	Yes

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CT417 Software Engineering III

Exam Duration: 2 Hours

Answer any **THREE** questions.

Question 1 [20 Marks]

Scenario: **SkyBankAI** is a mid-sized digital bank rapidly expanding its services through AI-driven financial tools, such as personalised investment advice and fraud detection. It operates across multiple regions and serves millions of customers through its mobile and web platforms. In order to remain competitive and responsive to user needs, SkyBankAI aims to release features weekly. However, their current software delivery workflow presents recurring challenges:

- Manual testing processes result in inconsistent test coverage, delays during high-traffic periods (e.g. end-of-quarter reporting), and occasional production bugs.
- Poor integration discipline among teams often leads to code conflicts and unstable releases.
- Frequent regressions occur due to the absence of automated regression tests and environment parity across staging and production.
- Security vulnerabilities, particularly those introduced through third-party libraries which were identified too late in the development cycle, risking compliance issues and user trust.

- a) During a recent deployment at SkyBankAI's GitHub Actions workflow failed at the **deployment stage**.

The developer received the following partial log:

```
Run docker pull ghcr.io/skybankai/app:latest
Error response from daemon: manifest for
ghcr.io/skybankai/app:latest not found: manifest unknown
Error: Process completed with exit code 1
```

Using only `git` and basic `docker` commands, show how you would:

- (1) Diagnose the issue,
- (2) Identify the cause, and
- (3) Rectify and redeploy.

[5 marks]

**Question 1 continued on next page.*

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- b) Design a simplified CI/CD pipeline using GitHub Actions for one of SkyBankAI's microservices built with Node.js.

The pipeline must:

- Run tests on pull requests
- Build and push a Docker image after merging into main
- Deploy to a staging server

List and explain the key steps in the pipeline.

[5 marks]

- c) A junior developer creates a pipeline where:

- Tests only run on the main branch.
- Every commit to main triggers a direct deployment to production.
- There is no rollback strategy.

Identify the risks in this design and propose improvements.

[5 marks]

- d) SkyBankAI's development teams follow a feature-based branching strategy and use GitHub (with GitHub Actions) for CI. However, during periods of high development activity, merge conflicts frequently occur when integrating feature branches into main.

Explain how merge conflicts typically arise in collaborative GitHub workflows, and describe a clear, practical process to prevent and resolve them.

[5 marks]

PTO

Question 2 [20 Marks]

Scenario: SkyBankAI is enhancing its software security posture and quality assurance by integrating automated testing tools into its CI/CD pipelines.

- a) Explain the key differences between Static Application Security Testing (SAST) and Dynamic Application Security Testing (DAST) by comparing:

- When they are applied in the software lifecycle
- What types of vulnerabilities they are suited to detect
- Whether they require access to source code
- A tool example for each

[5 marks]

- b) SkyBankAI has integrated OWASP ZAP into its pipeline, but the pipeline fails frequently, and no meaningful reports are generated.

Identify **two** likely technical causes for these failures and propose a practical resolution for each.

[5 marks]

- c) SonarQube static analysis is slowing down the CI/CD feedback loop. As a DevOps engineer, you are asked to optimise it without reducing scan quality.

Propose **two** changes to improve performance and explain the effect and trade-offs of each.

[5 marks]

- d) Despite having unit tests, SkyBankAI developers are concerned about low test coverage in the payment module.

Explain how a coverage tool (e.g., JaCoCo, Istanbul, or Coverage.py) can be used to identify low-coverage areas, and describe one strategy to increase coverage efficiently without slowing down the pipeline.

[5 marks]

PTO

Question 3 [20 Marks]

Scenario: As part of its accelerated development roadmap, SkyBankAI has integrated several third-party AI services and open-source packages into its core platform. While these tools increase development velocity and customer insight, they also introduce new **attack surfaces and security risks**, especially in the context of sensitive financial data and compliance standards.

As the Solution Architect, prepare an overview covering foundational concepts of vulnerability management and secure software development.

- a) Define the meaning of “vulnerability”. [1 mark]
- b) Define the meaning of “countermeasures”. [1 mark]
- c) What is zero-day vulnerability? [2 marks]
- d) SkyBankAI operates a 24/7 online banking platform monitored by its DevSecOps team. The following data was collected from three recent incidents over the past quarter:

#	MTTD	MTTR	Total Downtime	Notes
1	4 hours	2 hours	6 hours	Slow alert correlation
2	30 minutes	1 hour	1.5 hours	Automated alerting active
3	6 hours	4 hours	10 hours	Manual triage, patch delay

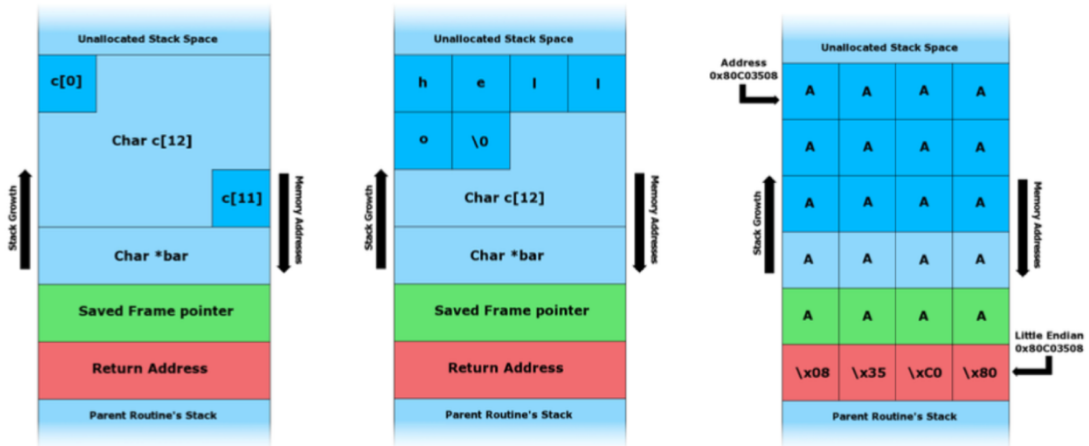
- Using the data above, calculate the average MTBF (Mean Time Between Failures) assuming that $MTBF = MTTD + MTTR$ for each incident. [4 marks]
- Identify which incident demonstrates the best operational performance and explain why. [2 marks]
- Recommend two practical improvements SkyBankAI could implement to reduce MTTD or MTTR in future incidents. [4 marks]

**Question 3 continued on next page.*

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e) Some library function in C are vulnerable to buffer overflow attack.

- Define and explain the concept of a “buffer overflow” and how this vulnerability can be exploited by attackers.



- Discuss TWO countermeasures that can be implemented to address the issue of “buffer overflow”.

[6 marks]

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Question 4 [20 Marks]

Scenario: SkyBankAI's backend system manages user authentication, account creation, payments, and notifications. As the system evolves, it has become increasingly difficult to extend and maintain due to:

- Repetitive logic for different user roles
- Expanding payment methods, each handled through conditional statements
- Hardcoded integration logic for notification services (e.g. email, SMS, push)
- Plans to introduce more user roles and asynchronous workflows.

As the technical design reviewer, you are tasked with proposing improvements using object-oriented principles and suitable design patterns.

a) Why is refactoring important for improving code reusability?

[4 marks]

b) Based on the scenario, identify **two** concrete design problems in the current system. For each:

- Describe the specific problem,
- Name the appropriate design pattern,
- Explain how it improves extensibility, maintainability, or modularity

[3 marks for each = 6 marks]

c) SkyBankAI currently uses logic like this:

```
public class UserController {
    public void authenticateUser(String role, String username,
        String password) {
        if (role.equals("Customer")) {
            // Authenticate customer
        } else if (role.equals("Manager")) {
            // Authenticate manager
        }
    }

    public void createUserAccount(String role, String username,
        String password, String extraInfo) {
        if (role.equals("Customer")) {
            // Customer account creation logic
        } else if (role.equals("Manager")) {
            // Manager account creation logic
        }
    }
}
```

**Question 4 continued on next page.*

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SkyBankAI plans to support additional user types (e.g. Partner, Auditor) and wants to avoid modifying this controller for each new role.

Refactor this design using:

- An abstract User class or interface.

[2 Marks]

- Role-specific subclasses (Customer, and Manager).

[4 marks]

- A UserFactory to handle user instantiation.

[4 marks]

END